

The Multiple Roles that IPv6 Addresses Can Play in Today’s Internet: Paper Review

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November 2025

1 The Problem

This paper asks how we should reinterpret IPv6 addresses now that we have an enormous address space and modern transport and security mechanisms. Instead of treating an IPv6 address as just “a bigger IPv4 address” (identifier + locator), the authors explore how multiple addresses and prefixes per host or network can encode semantics: privacy, load-balancing, multihoming, mobility, and service chaining.

It is not a full system paper. It is primarily an architectural / design paper with several concrete mechanisms and a small prototype (per-core addressing for a QUIC server).

2 Motivation

The motivation comes from two main trends. First, IPv6 is vastly under-utilized: only a vanishingly small fraction of the space is routed, yet we still behave as if addresses are scarce. Second, transport and security have moved up the stack: TLS provides endpoint identity and QUIC/MPTCP support migration and multipath. As a result, IP addresses no longer need to be stable identities.

This makes the question important: if we keep using IPv6 as “IPv4 with more bits,” we lose an opportunity to simplify multihoming and mobility, strengthen privacy, and improve performance. The paper does not prove these changes are necessary, but it shows that treating addresses as cheap opens up useful design space.

3 New/Key Ideas and Contributions

The key contribution is a catalog of ways one can exploit multiple IPv6 addresses and prefixes:

- **Multiple addresses per host:** Use short-lived privacy addresses together with multipath transports so flows can migrate as addresses rotate. This improves privacy and thwarts scanning without keeping old addresses alive for long.
- **Moving-target addresses for servers:** Servers can continually change their IPv6 address (e.g., derived from a secret) while multipath transports migrate existing connections. This generalizes prior NAT-based schemes.
- **Multi-addressed cellular hosts:** A mobile device can have different addresses for different radios or cells, using multipath to choose and switch paths, reducing the need for complex tunneling-based mobility.

- **Per-core (and per-process) addresses:** Instead of RSS hashing on a single IP, assign each core its own IPv6 address and steer packets by destination address. A prototype QUIC server with DPDK shows better load balance and up to $\approx 25\%$ throughput gains; extending this to per-process addresses strengthens isolation.
- **Host-based IPv6 multihoming:** Enterprises receive provider-aggregatable prefixes from each ISP and hosts get one address from each. With multipath TCP/QUIC, hosts can use and fail over between providers without the site announcing its own prefix. This could significantly reduce BGP table size and churn from multihomed stubs.
- **Multiple prefixes in the network:** Using SRv6 and multiple prefixes, operators can encode service chains, topology variants (low-latency vs low-cost), or multicast behavior directly in address structure.

Individually, many mechanisms build on prior work, but the paper’s novelty lies in systematically treating addresses as rich, expendable labels.

4 Main Findings

The quantitative results are modest but illustrative:

- **Per-core addressing:** a DPDK-based QUIC server using one IPv6 address per core achieves more balanced CPU utilization and up to $\approx 25\%$ higher throughput than an RSS-based baseline.
- **BGP impact:** analysis of BGP data shows that a large fraction of updates come from stub ASes; if those stubs used host-based multihoming instead of PI prefixes, much of this churn could disappear.

The rest of the paper’s value is conceptual rather than empirical.

5 Assumptions, Weaknesses, Limitations

Key assumptions include widespread deployment of IPv6, multipath transports, and hosts and middleboxes that can manage many addresses per interface. These are not fully true today, which makes deployment challenging.

Limitations and weaknesses:

- **Deployment complexity:** Host-based multihoming and cellular redesign require non-trivial changes to OSes, routers, and operational practice. Incremental deployment paths are only sketched.
- **Security/privacy analysis:** Moving-target addressing and per-connection addresses are argued to help privacy, but there is no rigorous analysis of new attack surfaces or correlation risks.
- **Ecosystem impact:** Ideas such as “no ports, only addresses” would break many existing tools and policies; compatibility costs are not fully addressed.

- **Limited evaluation:** Aside from the per-core experiment and some BGP statistics, most proposals remain at the design level.

Extensions could include full host-based multihoming implementations, detailed privacy/security studies of ephemeral addressing, and scalable NIC/OS support for per-core/process address assignment.

6 Lessons

The main lessons are:

- With IPv6, address space is no longer scarce; the bottleneck is operational complexity. We can afford to “spend” addresses to encode semantics.
- Because transport and identity have moved up the stack, IP addresses no longer need to be stable identities. This enables multi-address, short-lived, and role-specific addresses.
- Many long-standing problems (privacy, multihoming, mobility, load-balancing) can be re-framed as address-usage problems once we embrace multiple roles for IPv6 addresses.